Human Review: Null minimal displacement for almost nonexpansive maps

Original automatic proof: `solution_packet.pdf`
Checked date: 13 June 2026
Status: Manually checked.
Verdict: The proof appears correct.

Human Comments

The proof appears to give a correct answer to $(Q1)$. The main idea is to start from the Lin–Sternfeld theorem, which gives a Lipschitz map

$$F : K \to K$$

with positive minimal displacement. In particular, F has no fixed point. The solution then modifies F by damping the displacement $F(x) - x$ with a positive Lipschitz scalar $\lambda(x)$ whose infimum on K is zero.

The resulting map has the form

$$G_\varepsilon(x) = x + c\lambda(x)(F(x) - x) = (1 - c\lambda(x))x + c\lambda(x)F(x).$$

Choosing $c > 0$ sufficiently small makes G_ε $(1+\varepsilon)$-Lipschitz, while the fact that $\lambda(x) > 0$ for every $x \in K$ ensures that no fixed point is introduced.

Verification of the Construction

The scalar function λ is built from a separated sequence $(x_n) \subset K$. One first defines values tending to zero, for instance $\lambda(x_n) = 1/n$, and extends them Lipschitzly to K, truncating if necessary so that $0 < \lambda \le 1$. The important features are:

$$\lambda(x) > 0 \quad \text{for all } x \in K, \qquad \inf_K \lambda = 0.$$

These two facts pull in opposite directions in exactly the way needed here: pointwise positivity preserves the fixed-point-free property, while the vanishing infimum forces the minimal displacement of the new map to be zero.

Indeed, since K is convex and $0 < c\lambda(x) \le 1$, the expression

$$G_\varepsilon(x) = (1 - c\lambda(x))x + c\lambda(x)F(x)$$

lies in K. If $G_\varepsilon(x) = x$, then

$$c\lambda(x)(F(x) - x) = 0.$$

Since $c > 0$ and $\lambda(x) > 0$, this would imply $F(x) = x$, contradicting the positive minimal displacement of F.

On the other hand, along the chosen separated sequence,

$$\|G_\varepsilon(x_n) - x_n\| = c\lambda(x_n)\|F(x_n) - x_n\| \le \frac{c\ \mathrm{diam}(K)}{n} \longrightarrow 0.$$

Thus

$$d(G_\varepsilon, K) = 0.$$

Lipschitz Estimate

The Lipschitz estimate also looks correct. If M is a Lipschitz constant for F, $D = \operatorname{diam}(K)$, and L_λ is a Lipschitz constant for λ, then $H = F - I$ is $(M+1)$-Lipschitz and bounded by D. Hence

$$\begin{aligned} \|\lambda(x)H(x) - \lambda(y)H(y)\| &\le \|H(x) - H(y)\| + |\lambda(x) - \lambda(y)|\,\|H(y)\| \\ &\le \big((M+1) + DL_\lambda\big)\|x - y\|. \end{aligned}$$

Taking c small enough therefore gives

$$\operatorname{Lip}(G_\varepsilon) \le 1 + \varepsilon.$$

Review Outcome

Archive this packet as an apparently correct full solution to $(Q1)$. The result seems stronger than what was asked, since it gives the $(1+\varepsilon)$-Lipschitz and null-minimal-displacement conclusion for every preassigned noncompact bounded closed convex K, rather than merely producing some suitable set K in a given space.

I recommend checking with the authors to confirm that this matches their intended formulation and to get their opinion on the mathematical value and novelty of the observation.